

JAI VARDHAN SINGH

Gameplay Programmer · Unreal Engine 5 Developer · Solo Indie Game Developer

Delhi, India · mrcingstudio@gmail.com · +91 9389756180 · [Portfolio](#) · [Fab](#) · [LinkedIn](#)

PROFESSIONAL SUMMARY

Self-taught Unreal Engine 5 gameplay programmer and solo indie developer with 5+ years in-engine. Built a 60-level mobile action-puzzle game end-to-end — gameplay systems, AI, UI, optimization, monetization, and store submission — now in closed testing on Google Play. Shipped a tactical FPS gameplay framework and weapon asset packs generating revenue on Epic's Fab marketplace, and prototyped a story-driven FPS in 8 days for a game jam. Specialist in Blueprint architecture, gameplay mechanics, mobile optimization (Android), procedural generation, and rapid prototyping. Full-pipeline owner comfortable shipping under pressure.

CORE & TECHNICAL SKILLS

Gameplay Programming · Game Systems Design · Unreal Engine 5 / 4 · Blueprint Scripting (expert) · Rapid Prototyping · Mobile Optimization (Android) · Performance Profiling & Debugging · Procedural Generation · AI (enemy behaviour, companion AI, perception & audio detection) · Combat & Weapon Systems · Inventory, Save & Progression Systems · Physics · UMG / UI · Niagara VFX · Animation Blueprints · Blender (character modeling, rigging, animation, UV, materials, texturing) · Git / GitHub · Visual Studio · Google Play publishing · Fab marketplace publishing

PROJECTS (PROFESSIONAL EXPERIENCE)

THE RUFFIES — Solo Developer | UE5 · Android · Closed Testing on Google Play

- Built a 60-level action-puzzle adventure 100% solo: gameplay programming, AI, UI, characters, animation, audio, VFX, monetization, and store submission.
- **Designed a procedural maze generation system** — stack-based depth-first search over a 2D grid — powering all 60 optimized level layouts.
- **Engineered a runtime HISM coin-replacement system**, converting thousands of individual actors into batched instances — major draw-call reduction and FPS gains on low-end Android devices.
- Cut Android AAB size from 280MB to under 200MB; shipped rewarded ads, banner ads, and in-app purchases through Google Play policy review.
- Implemented persistent save system, character progression, shop, multiple playable characters, and upgradeable abilities (Magnet, Shield, Spring Jump, Speed Boost).

TACTICAL WARFARE KIT — Solo Developer | UE5 · Released on Fab (Epic Games Marketplace)

- Modular tactical FPS gameplay framework — originally a full game, re-architected into a documented, revenue-generating marketplace product.
- Weapon systems (inventory, switching, knife, grenades, recoil, weapon sway); enemy AI with perception and audio detection; companion AI and squad communication.
- Scalable architecture designed for other developers to extend; roadmap includes multiplayer via Epic Online Services.

LOOSE END — Solo Developer | UE5 · PC · 8-Day Game Jam Build (KGeN Road To Game Jam)

- Cinematic first-person action prototype: protect an NPC physically tethered to the player by a rope, across four story-driven missions.
- Movement stack (wall running, sliding, rocket jump, grappling, dash kill), rope-constrained companion physics, AI combat, shop and weapon unlocks — scoped, built, and delivered solo in 8 days.

Additional: Two published Fab asset packs — Side Arm Steel (pistols: modeling, reload/fire animations, Blueprint integration) and Xeno Fang & Phantom Slice (stylized sci-fi melee, materials/texturing, game-ready). Two mobile titles in development (parkour runner; physics-perspective puzzle).

WORK EXPERIENCE

Security Associate — Indira Gandhi International Airport, Delhi | Dial Gmr · 2024–Present

- Full-time role in a high-pressure, zero-error operational environment: strict procedural compliance, incident response, teamwork, and clear communication — while independently building and shipping game projects.

EDUCATION

BBA, Aviation Management — Galgotias University 2021-2024